Mapping the land-use footprint of Brazilian soy embodied in international consumption: a spatially explicit input–output approach

Liza Buriak[1*]

[1*]Institute for Ecological Economics, Vienna University of Economics and Business (WU), Vienna, Austria.

Corresponding author(s). E-mail(s): TODO@wu.ac.at;

Abstract

TODO — Data-only working document. The full abstract lives in the main manuscript. Placeholder so the file compiles standalone.

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1 Data

The model is built entirely from publicly and freely available data. Inputs fall into four groups: (i) subnational Brazilian statistics on soy production, processing, domestic use and trade; (ii) spatial layers describing the multimodal transport network; (iii) administrative and environmental geographies used to place and attribute the footprint; and (iv) the global input–output backends (FABIO and EXIOBASE) plus the independent supply-chain benchmark (Trase). Unless noted otherwise, all series are annual and cover 2010–2020 at the Brazilian municipality (*município*) level (∼5,570 units). Where a dataset is provided only for a single reference year (e.g. gridded livestock, household consumption), that year is stated and the value is held constant, as detailed below. All source URLs were last accessed in July 2026. A programmatic downloader for the machine-retrievable sources is provided with the code (`code/download_data.py`).

1.1 Subnational production, processing, consumption and trade

1.1.1 Agricultural production and population (IBGE)

Municipal soybean production, planted area and harvested area come from the Brazilian Institute of Geography and Statistics (IBGE) automated recovery system SIDRA, table 1612 (*Produção Agrícola Municipal*), variables 109 (planted area, ha), 216 (harvested area, ha) and 214 (production, tonnes), for the soybean class `c81/2713`. Data are pulled from the SIDRA API (https://apisidra.ibge.gov.br/values/t/1612/n6/all/v/109,216,214/p/{year}/c81/2713; table page https://sidra.ibge.gov.br/tabela/1612). Resident population by municipality is taken from SIDRA table 6579 (variable 9324), which runs through 2021; for 2022 the 2022 Census table 9514 is used. Population enters as a proxy for household soy-oil consumption (Methods).

1.1.2 Livestock herds (IBGE)

Municipal herd sizes for ten animal categories (cattle, buffalo, horses, pigs and breeding sows, goats, sheep, chickens and laying hens, quail) are from SIDRA table 3939 (*Efetivo dos rebanhos*, variable 105, herd-type classification `c79`). Milked-cow counts, used to split dairy from meat herds, are from SIDRA table 94 (variable 106). Feedlot cattle come from the IBGE Agricultural Census: table 919 (2006 Census) extrapolated with the ABIEC growth rate, and table 6911 (2017 Census). All are retrieved from the same SIDRA API endpoint pattern (https://apisidra.ibge.gov.br/values/t/{table}/n6/all/...).

1.1.3 Soy crushing and refining capacity (ABIOVE)

Installed processing capacity — crushing and refining/bottling facilities with active-facility counts and daily capacities by municipality — is from the Brazilian Association of Vegetable Oil Industries (ABIOVE), "Capacidade Instalada" statistics (https://abiove.org.br/estatisticas/). These spreadsheets are downloaded manually and stored as `Processing_facilities_{year}_ABIOVE.xlsx` (sheets `processing_MUN` and `refining_bottling_MUN`).

1.1.4 Biodiesel capacity (ANP)

Biodiesel plant capacity (m^3/day per municipality) and the regional share of soy as biodiesel feedstock are from the National Agency of Petroleum, Natural Gas and Biofuels (ANP) Statistical Yearbook, Table 2.6 (https://www.gov.br/anp/pt-br/centrais-de-conteudo/publicacoes/anuario-estatistico/), stored as `Biodiesel_capacity_{year}_ANP.xlsx`.

1.1.5 Household soy-oil consumption and grain storage (IBGE)

Per-capita soy-oil acquisition by state, used to distribute national food-oil use across municipalities, is from the IBGE Household Budget Survey (POF) 2017–2018 — the most recent available — held constant across years (https://www.ibge.gov.br/estatisticas/sociais/educacao/9050-pesquisa-de-orcamentos-familiares.html). Grain-storage capacity by municipality, used to allocate stock changes, is from the IBGE

Stocks Survey (*Pesquisa de Estoques*), SIDRA table 278 (https://sidra.ibge.gov.br/tabela/278).

1.1.6 National commodity balances (FAO)

National supply–use balances that the municipal totals are reconciled against are the FAO Food/Commodity Balance Sheets for Brazil (area code 21), items 2555 (soybeans), 2571 (soybean oil) and 2590 (soybean cake), covering production, imports, exports, food, feed, seed, processing, other uses and stock variation. These are extracted from the FAOSTAT bulk download (https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_All_Data_(Normalized).zip; domain page https://www.fao.org/faostat/en/#data/FBS).

1.1.7 Livestock distribution and feed rates (FAO)

Herds are disaggregated into production systems using the FAO Gridded Livestock of the World (GLW3, 2010 reference year) density rasters for cattle, buffalo, pigs (extensive/intensive/industrial) and chickens (extensive/intensive) [1], distributed via the Harvard Dataverse (https://dataverse.harvard.edu/dataverse/glw), together with the FAO GLEAM ruminant production-system classification (`glps_gleam_61113_10km.tif`) and GLEAM per-animal feed ratios (dry-matter intake and soy/soy-cake shares) [2], https://www.fao.org/gleam/resources/en/. These are static 2010 layers, reused for all years; the resulting production-system *shares* (not herd sizes, which are annual) are therefore held constant over the study period.

1.1.8 Municipal-level trade (COMEX/MDIC)

Municipality-level exports and imports are from the Brazilian foreign-trade database COMEX Stat (Ministry of Development, Industry, Trade and Services), filtered to the soy HS4 codes 1201 (soybeans), 1507 (soybean oil) and 2304 (soybean cake). Per-year municipal files and the country and municipality lookup tables are downloaded from https://balanca.economia.gov.br/balanca/bd/comexstat-bd/mun/ (files `EXP_{year}_MUN.csv`, `IMP_{year}_MUN.csv`) and https://balanca.economia.gov.br/balanca/bd/tabelas/ (`PAIS.csv`, `UF_MUN.csv`).

1.2 Transport network

The subnational routing of physical flows (Methods) uses four multimodal network layers:

- **Roads.** OpenStreetMap road network (`gis_osm_roads_free_1.shp`), obtained as a Brazil extract from Geofabrik (https://download.geofabrik.de/south-america/brazil.html); © OpenStreetMap contributors.
- **Waterways.** Federal inland-waterway corridors (*Hidrovias*) from the National Department of Transport Infrastructure (DNIT), https://www.gov.br/dnit/pt-br.
- **Railways.** Rail lines, stations and annual rail-cargo flows by commodity (soybean, soy cake, vegetable oil) from the National Land

Transport Agency (ANTT), 2006–2021 (https://dados.antt.gov.br/dataset/transporte-ferroviario-de-cargas-e-passageiros).

- **Ports and port cargo.** Port locations and annual waterway-cargo records from the National Waterway Transport Agency (ANTAQ) statistical system, 2017–2022 with a 2013 fallback for earlier years (https://web.antaq.gov.br/anuario/).

A small number of soy-relevant rail stations and supplementary port points were hand-curated and are distributed with the code.

1.3 Administrative and environmental geographies

Boundaries. Municipal polygons are IBGE administrative meshes, retrieved with the `geobr` R package [3] (https://ipeagit.github.io/geobr/; also on CRAN, https://cran.r-project.org/package=geobr) and municipality code/name lookups from the IBGE localities API (https://servicodados.ibge.gov.br/api/v1/localidades/municipios); state-level boundaries for mapping use GADM v3.6 (`gadm36_BRA_1`, https://gadm.org/).

Biomes and frontier. Footprint origin is attributed to biomes using the IBGE 1:250,000 biome layer (`lm_bioma_250.shp`: Amazônia, Cerrado, Caatinga, Mata Atlântica, Pantanal, Pampa; https://www.ibge.gov.br/geociencias/informacoes-ambientais/estudos-ambientais/15842-biomas.html). The agricultural *frontier* is defined as the Amazon biome plus MATOPIBA, the Cerrado expansion zone spanning Maranhão, Tocantins, Piauí and Bahia. MATOPIBA is operationalized as the intersection of the Cerrado biome with these four states, which closely approximates the official Embrapa delimitation of the region [4].

Grid refinement (optional). An optional 30 m refinement of the municipal footprint uses MapBiomas annual soy land-use tiles [5] (https://mapbiomas.org/), exported from Google Earth Engine with an adapted version of the official MapBiomas user toolkit (https://github.com/mapbiomas-brazil/user-toolkit). This layer is a visual refinement and is not load-bearing for the headline results.

1.4 Global supply-chain models and benchmark

FABIO. Subnational flows are nested into the Food and Agriculture Biomass Input–Output model, version 2 (FABIO v2), built by the fineprint team (https://github.com/fineprint-global/fabio). The v2 matrices cover 2010–2023 at a resolution of 22,263 processes with 123 environmental extensions (files `E`, `Z_mass_b`, `Z_value_b`, `Y_b`), complemented by the v2 build ingredients (balanced commodity balance sheets, bilateral trade, tidied livestock and price tables) at the v2 classification (181 regions × 123 items). The published FABIO v1.1 release (1986–2013; DOI `10.5281/zenodo.2577067`) is cited for provenance only and is *not* used, as its classification and time span differ [6].

EXIOBASE. The non-soy "background" of the economy is supplied by EXIOBASE 3 [7] in the product-by-product (pxp) transformation, 2000–2020, with 9,800 × 9,800 flow matrices (49 regions × 200 products) and their environmental satellites (https://www.exiobase.eu/; data on Zenodo, DOI `10.5281/zenodo.3583070`). These are hybridised with FABIO for the non-soy quadrant of the footprint model.

Bilateral trade (FAO). Re-export disentangling and the FABIO trade layer draw on the FAOSTAT Detailed Trade Matrix for Brazil (items 236 soybeans, 237 soybean oil, 238 soybean cake), from the FAOSTAT bulk download (https://bulks-faostat.fao.org/production/Trade_DetailedTradeMatrix_E_All_Data_(Normalized).zip).

Independent benchmark (Trase). Model flows from origin municipality to country of first import are validated against the Trase Brazil soy composite dataset, version 2.6.1 (`brazil_soy_v2_6_1_composite.csv`, 2004–2022), from the Trase initiative (https://trase.earth/) [8, 9]. Trase is used exclusively as a validation benchmark; no Trase data enter the model itself.

1.5 Coverage and summary

The binding constraint on temporal coverage is FABIO v2 (2010–2023), so the footprint series is reported for the decade 2010–2020; the origin-to-importer flow benchmark against Trase extends back to 2004 and supports a longer validation backdrop. Table 1 summarises all inputs.

1.6 Data availability

All input data used in this study are publicly and freely available from the providers listed above. They are not redistributed with the code because of differing licences; a download script with direct source links is provided (`code/download_data.py`), and the datasets requiring manual retrieval (ABIOVE, ANP, POF, MapBiomas tiles) are documented with step-by-step instructions in the repository. The intermediate and final model outputs underlying the figures will be deposited on Zenodo upon publication.

1.7 Code availability

The complete modelling pipeline (data preparation, transport allocation, FABIO/EXIOBASE nesting, footprint accounting, validation and figures) is written in R and Python and released under the GNU GPL v3 at `<repository URL --- TODO>`. The results reported here have no proprietary software dependency. An experimental multimodal-transport module included in the repository (not used for the reported results) additionally requires a GAMS licence.

References

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Table 1 Input datasets, by role. All are publicly and freely available; per-source access details and URLs are given in the text. "Munis" = Brazilian municipalities.

Dataset	Provider	Role	Coverage
Soy production, area (t.1612)	IBGE (SIDRA)	production	munis, annual
Population (t.6579/9514)	IBGE (SIDRA)	food-use proxy	munis, annual
Herds (t.3939), milk cows (t.94)	IBGE (SIDRA)	feed demand	munis, annual
Feedlot cattle (t.919/6911)	IBGE (Census)	feed demand	munis, 2006/2017
Crush/refining capacity	ABIOVE	processing	munis, annual
Biodiesel capacity	ANP	other (biodiesel) use	munis, annual
Soy-oil acquisition (POF)	IBGE	food-oil allocation	states, 2017–18
Grain storage (t.278)	IBGE	stock allocation	munis, annual
Commodity balance (FBS)	FAO (FAOSTAT)	national reconciliation	Brazil, annual
Gridded livestock (GLW3)	FAO (Dataverse)	system split	10 km, 2010
Production systems, feed (GLEAM)	FAO	feed rates	10 km, static
Municipal trade	COMEX/MDIC	exports/imports	munis, annual
Road network	OSM (Geofabrik)	transport	national
Waterways	DNIT	transport	national
Rail lines, stations, cargo	ANTT	transport	national, 2006–21
Ports, port cargo	ANTAQ	transport	national, 2017–22
Municipal boundaries	IBGE (geobr)	geometry	munis
State boundaries	GADM v3.6	mapping	states
Biomes 1:250k	IBGE	origin attribution	national
Soy land tiles	MapBiomas (GEE)	grid refinement	30 m, annual
FABIO v2 (MRIO)	fineprint	supply-chain nesting	global, 2010–23
EXIOBASE 3 pxp	EXIOBASE	non-soy background	global, 2000–20
Detailed trade matrix	FAO (FAOSTAT)	re-exports/trade	Brazil, annual
Trase composite v2.6.1	Trase	validation benchmark	munis, 2004–22

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